

London Tube Station

Recommendation

Susquehanna Datathon Submission

by
Matthew Ryan

LSESU Mathematics $\times$ LSESU Data Science and Quantitative

Bachelor of Science
in
Mathematics with Data Science

London School of Economics
London, United Kingdom



© 2026
Matthew Ryan

1 Introduction

London's Underground network, despite being one of the most extensive metro systems in the world, leaves significant gaps in coverage across the city. South-east London in particular remains structurally disconnected from the Underground network, with large residential populations relying entirely on national rail services and buses for connectivity to central London. This report presents a data-driven recommendation for the location of a new London Underground station, as part of an advisory service to Transport for London (TfL).

Using a scoring framework applied across all 4,994 Lower Layer Super Output Areas (LSOAs) in London, this report identifies Orpington in the London Borough of Bromley as the optimal location for a new station. The proposition is to extend the Bakerloo line through south-east London to Orpington, connecting one of the most underserved areas of the city to the wider Underground network.

2 Methodology

2.1 Overview

A quantitative scoring framework was created to evaluate all 4,994 LSOAs in London across eight variables, each capturing a distinct dimension of transport need and demand. LSOAs are the standard unit of geographic analysis used by the Office for National Statistics (ONS), comprising approximately 1,500 residents each, providing sufficient granularity to identify specific locations within London while being consistently available in public datasets.

2.2 Data Preprocessing

All variables were sourced from publicly available datasets and APIs, and joined on the LSOA code for each area, to create a full DataFrame with a unique identifier. Some variables, like `population_growth`, were only available at the borough level, and were therefore extrapolated. Other datasets had the old 2011 LSOA codes and didn't account for the new LSOAs that have been created since then; these also had to be extrapolated from the borough medians.

2.3 Scoring Framework

Three scoring schemes were applied to the data, each reflecting a different set of priorities with the weighting of variables:

Ridership-Focused: Designed to favour variables that would lead to an increase in ridership on the London Underground.

Equity-Focused: Designed to favour more deprived areas in London, that may benefit from better public transport more.

Balanced: Assigns equal weight (15%) to all primary variables, treating ridership potential and equity as equally valid criteria.

Each LSOA was assigned a composite score under each scheme as a weighted sum of normalised variable values. Rankings were then derived from these scores and combined using a Borda count to produce a final overall ranking. This meant a candidate that performed highly across all priorities was chosen.

2.4 Sensitivity Analysis

The use of three distinct weighting schemes constitutes a sensitivity analysis of the scoring framework. A candidate that ranks highly across all three schemes provides stronger evidence of genuine underservice than one that only scores well under a single set of assumptions. The Borda count aggregation is a well-established method for combining rankings and is used here to surface the most consistently high-performing candidates.

2.5 Geographical Analysis

Once the most suitable LSOA was identified, geographical analysis by inspection was done on the surrounding area, to pick a suitable location for an underground station. As well as this, ridership at the proposed station was estimated by looking at the population of the surrounding catchment area, which was an area of LSOAs that were less than or equal to 3.0km away from the new station.

3 Data and Variables

3.1 DataFrame Schema

Variable	Source	Description
lsoa_code*	ONS Census 2021	The LSOA code of the area. Unique Identifier.
borough	ONS Census 2021	The name of the London Borough of the LSOA.
population_density	ONS Census 2021	The number of residents per hectare in the LSOA.
population_growth	London Datastore	Projected borough-level growth rate from 2025–2040.
nearest_station	TFL StopPoint API	Straight-line distance to nearest London Underground station.
nearest_crowding	TFL Station Usage	Average entries into nearest London Underground Station (Tue - Thu).
commuter_outflow	ONS Census 2021	Proportion of commuters in the area travelling by car or bus.
deprivation_score	IMD 2019	The index of Multiple Deprivation score.
ptal_score	London Datastore 2015	Public transport accessibility level in the LSOA.
airbnb_count	Inside AirBnB 2025	AirBnB listing count per LSOA.

The most difficult and time-consuming part of the analysis was obtaining all of the data in this schema. Getting the data into a structured format like this was essential to be able to have explainability and interpretation with the scoring system, which otherwise would've been extremely difficult with the initial unstructured data.

3.2 Exploratory Data Analysis

After getting the data into the correct format, visualisations were used to identify any discrepancies that may skew the scoring frameworks. In particular, any distributions with heavy skews or correlated variables needed to be dealt with to avoid influencing the scoring systems heavily in a particular direction.



As seen on the distribution plots, `population_density`, `nearest_station`, `nearest_crowding` and `airbnb_count` were all heavily skewed to the right, so log transformations ($\log(1 + x)$) were applied to these. It was also noted that `population_growth` was also heavily skewed, but no transformation was applied here as the values were already extrapolated from the borough level.

The correlation matrix shows that none of the variables were correlated with $r > 0.70$, which meant that all correlations were deemed as sufficiently independent to avoid influencing the scoring frameworks too heavily.

3.3 Normalisation

All variables were min-max normalised to a $[0, 1]$ scale to ensure comparability across variables with different units and magnitudes:

$$x_{\text{normalised}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Variables where a higher raw value indicates better existing connectivity - specifically PTAL score - were inverted prior to normalisation so that higher normalised values consistently represent greater need across all variables.

4 Scoring and Analysis

4.1 Scoring Framework Weighting

Three scoring schemes were applied to the normalised data, each reflecting a different set of priorities:

Variable	Ridership	Equity	Balanced
Population Density	25%	15%	15%
Commuter Outflow	25%	10%	15%
Nearest Crowding	20%	0%	15%
Nearest Station	15%	15%	15%
PTAL Score	10%	25%	15%
Deprivation Score	0%	30%	15%
Population Growth	5%	5%	5%
AirBnB Count	0%	0%	5%

Variable weights under each scoring scheme

4.2 Sensitivity Analysis

Rankings were derived from each framework's composite scores and combined using a Borda count to produce a final overall ranking. This approach ensures the recommendation is robust to different weighting assumptions, and provides flexibility as candidates which are stronger when prioritising different variables can be identified.

LSOA	Borough	Ridership	Equity	Balanced	Overall
E01000739	Bromley	2	5	1	1
E01000789	Bromley	1	10	2	2
E01000729	Bromley	6	8	4	3
E01000721	Bromley	15	4	3	4
E01001055	Croydon	27	13	18	5
E01000730	Bromley	29	26	7	6
E01000724	Bromley	17	37	13	7
E01001602	Greenwich	3	61	6	8
E01001603	Greenwich	25	39	9	9
E01001080	Croydon	75	1	5	10

Top 10 LSOAs by Borda count overall rank

The results consistently identify South-East London as the most underserved region of the network. LSOA E01000739 in Bromley ranks first overall.

5 Candidate Selection

5.1 Selected Candidate

Based on the Borda count ranking, LSOA E01000739 in the London Borough of Bromley is selected as the proposed location for the new station. The values and normalised score for this LSOA are shown:

Variable	Raw Value	Normalised Score
population_density	77.1	0.550
population_growth	7.5%	0.087
nearest_station	13.1km	0.856
nearest_crowding	35,456	0.769
commuter_outflow	52.5%	0.876
deprivation_score	42.4	0.643
ptal_score (inverted)	115.5	0.949
airbnb_count	4	0.247

- **Connectivity:** Near-worst public transport accessibility in London, scoring in the 95th percentile for poor PTAL (normalised score: 0.949).
- **Suppressed Demand:** 52.5% of active commuters travel by car or bus, one of the highest rates across all London LSOAs, indicating significant latent ridership potential.
- **Coverage Gap:** 13.1km from the nearest tube station, hugely above the London median of 1.3km.

5.2 Geographic Analysis

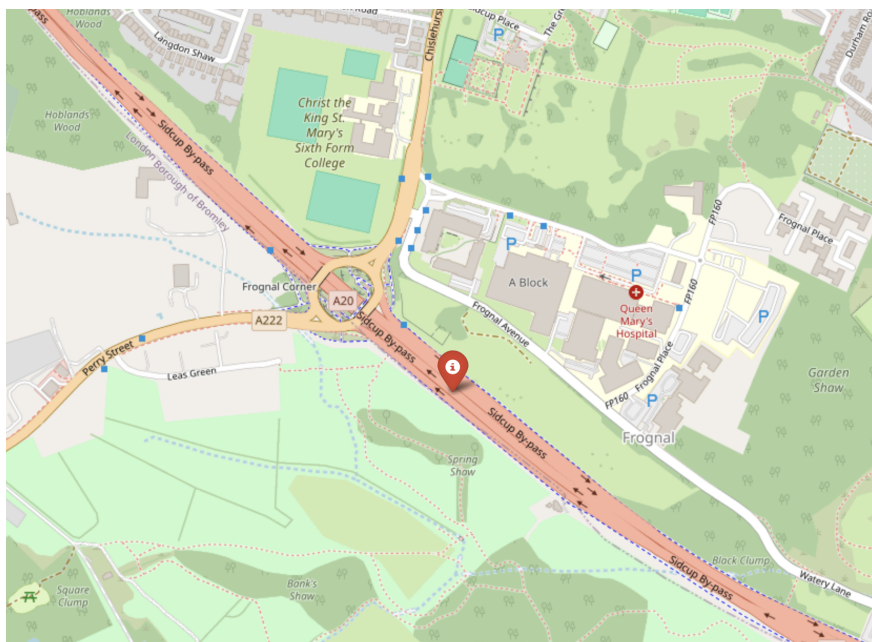
The proposed station is located in Orpington, in the London Borough of Bromley, on the south-eastern edge of the Greater London built-up area. Orpington is identified in the London Plan as one of 35 major centres in Greater London, reflecting its significance as a suburban economic and residential hub.

Despite this designation, the area is heavily car-dependent. Residents of the BR5 postcode average 2.19 cars per household, significantly above the UK average of 1.4 - consistent with the LSOA's commuter outflow score of 52.5%, one of the highest in London. Existing public transport consists of Southeastern national rail services to Charing Cross and Cannon Street via Grove Park, and Victoria via Bromley South,

alongside numerous bus routes. However, these services require interchange to access the wider tube network and do not provide the direct, flexible connectivity that Underground access would offer.

Whilst the LSOA centroid used in the spatial analysis is located at 51.401234°N, 0.105364°E - in the middle of a housing estate - a more suitable construction site has been identified through inspection at **51.417601°N, 0.100140°E**. This location offers three key practical advantages:

- Adjacent to a Sidcup By-Pass Road, providing accessible passenger entry and construction vehicle access
- In close proximity to Queen Mary's Hospital, serving both staff and patients who currently rely on car or bus travel
- Sufficient open land in Scadbury Park available for construction, consistent with the problem statement assumption that TfL has the administrative power to repurpose land



Proposed station location (red marker) at 51.417601°N, 0.100140°E, adjacent to Queen Mary's Hospital and Sidcup bypass. Proposed construction site at Scadbury Park is visible to the south.

5.3 Line Routing

The proposed station would be served by an extension of the Bakerloo line, consistent with TfL's own long-term planning for south-east London. With TfL's plans to extend the Bakerloo Line (TfL: 'Bakerloo Line Extension'). TfL has been consulting since 2014 on extending the Bakerloo line from Elephant Castle to Lewisham, with proposals for a further extension to Hayes and Beckenham Junction and potentially as far as Bromley. Public support for the extension is strong, with 76% of residents and 89% of businesses along the proposed route in favour.

The proposed station at Orpington would represent a logical terminus on this extended route connecting residents directly to central London via Lewisham, Elephant Castle, Waterloo, and Oxford Circus without interchange. This would transform the commuting options available to residents who currently rely on Southeastern national rail services requiring interchange at Grove Park or Bromley South.

The alignment of the data-driven recommendation with TfL's own planning framework serves as independent validation of the methodology - the scoring model, constructed without reference to existing proposals, identified the same geographic corridor as TfL's transport planners.

5.4 Ridership Estimate

Based on the sum of the populations of the LSOAs that are within 3.0km of the proposed station location, the station would be within close proximity to approximately 76883 people. We've already established that around 50% of commuters in the area commute by car or bus, so we can assume that 25% of the population would make use of this new station, at approximately 19,000 people being served. With the station giving access to Queen Mary's Hospital, this figure could be even higher.

6 Limitations

6.1 Data Currency

Several datasets used in this analysis used the 2011 Census LSOA boundaries rather than the 2021 Census boundaries. The Index of Multiple Deprivation (IMD) was last published in 2019, and PTAL scores are derived from a 2015 dataset. Whilst these remain the most recent publicly available datasets at LSOA level, some newly created

LSOAs had to be extrapolated from the medians of the borough values. Population growth projections are derived at borough level and applied uniformly to constituent LSOAs, which does not account for variation in development and housing growth.

6.2 External Rail Networks

The the analysis considers only London Underground stations. National Rail and Overground services are excluded from the distance to nearest station calculations. Whilst the nearest Thameslink station to the proposed location is approximately 3.7km away from the chosen candidate and there are southeastern rail services available nearby, the analysis is still valid as London Underground offers much more flexibility. However, a more rigorous analysis would implement other forms of rail services.

6.3 Scoring Framework

The weighted scoring framework relies on subjective weight assignments that, whilst justified through three distinct schemes and validated through sensitivity analysis, remain inherently arbitrary. The Borda count aggregation assumes equal importance of all three schemes, which may not reflect TfL's actual priorities. Furthermore, variables such as population density and PTAL are moderately correlated, introducing a degree of double-counting in the composite score despite no pair of variables exceeding a correlation of 0.7.

6.4 Ridership Estimate

The ridership estimate was the hardest to calculate. Since the nearest tube station is more than 13km away and in a more densely populated area of London, it was hard to compare other London Underground stations. Ridership estimate was calculated by finding the population of a catchment area of LSOAs that are within 3.0km of the proposed station, and estimating that the station would serve 25% of the population in this catchment area at 19,000 people. Whilst this was an assumption, under less strict time constraints there may have been a better way to quantitatively justify this percentage.

References

- [1] Office for National Statistics (2021). *2021 Census: LSOA Population Estimates*. Available at: <https://www.ons.gov.uk>
- [2] Office for National Statistics (2021). *LSOA December 2021 Boundaries Generalised Clipped EW BGC*. Available at: <https://geoportal.statistics.gov.uk>
- [3] Office for National Statistics (2021). *LSOA Area Measurements*. Available at: <https://geoportal.statistics.gov.uk>
- [4] Office for National Statistics (2021). *Census 2021: Method of Travel to Workplace (TS061)*. Available at: <https://www.ons.gov.uk>
- [5] Office for National Statistics (2021). *Output Area to LSOA to MSOA December 2021 Lookup*. Available at: <https://geoportal.statistics.gov.uk>
- [6] London Datastore. *Land Area and Population Density*. Available at: <https://data.london.gov.uk>
- [7] Transport for London (2024). *TFL StopPoint API*. Available at: <https://api.tfl.gov.uk>
- [8] Transport for London (2024). *Station Usage Data: Annual Entry and Exit Figures*. Available at: <https://tfl.gov.uk/info-for/open-data-users>
- [9] Transport for London (2015). *Public Transport Accessibility Levels (PTAL) by LSOA*. London Datastore. Available at: <https://data.london.gov.uk>
- [10] Ministry of Housing, Communities and Local Government (2019). *English Indices of Deprivation 2019*. Available at: <https://www.gov.uk/government/statistics/english-indices-of-deprivation-2019>

-
- [11] Inside AirBnB (2025). *London Listings Dataset, September 2025*. Available at: <https://insideairbnb.com/get-the-data>
 - [12] Transport for London (2024). *Bakerloo Line Extension*. Available at: <https://tfl.gov.uk/corporate/about-tfl/how-we-work/planning-for-the-future/bakerloo-line-extension>
 - [13] Lewisham Council (2024). *Overwhelming London-wide Support for Bakerloo Line Upgrade and Extension*. Available at: <https://lewisham.gov.uk>
 - [14] Wikipedia (2025). *Orpington*. Available at: <https://en.wikipedia.org/wiki/Orpington>
 - [15] Greater London Authority (2021). *The London Plan 2021*. Available at: <https://www.london.gov.uk/programmes-strategies/planning/london-plan>